

Boltzmann Transport from Observer-Patch Consistency: a Theorem-Conditional Reduction

OPH cosmology program

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Abstract

This note derives the general-relativistic photon–baryon Boltzmann transport used in CMB computations *inside* the OPH branch, as a theorem-conditional reduction: given the Einstein-branch geometry theorem, the recovered U(1) electromagnetic sector, and the reversible repair-register structure, the observer-facing photon record dynamics obeys the standard Liouville equation with the Thomson collision functional, and its linearization on the capacity-closure FRW background is exactly the standard multipole hierarchy. OPH does not modify radiative transfer at first order; it *inherits* it. The value of the derivation is provenance: it identifies precisely (i) which OPH theorems supply each ingredient, (ii) the two places where OPH-specific kernels could enter (a number-changing repair-exchange channel Γ_{rec} and a collar-packet stress B_A), and (iii) the irreducible imports (absolute electron mass, recombination history) that any honest physical-CMB receipt must declare. One new finite-lattice receipt (BOLTZ-R1, refinement Liouville) is defined and is testable in the public simulator. This note does not close the physical CMB prediction gate; it reduces that gate to a finite, named list.

1 Claim boundary

Everything below is conditional on the branch theorems it cites. The derivation upgrades the status of the *transport form* used by the diagnostic CAMB lane from “imported wholesale” to “theorem-conditional reduction with declared imports.” It does not certify the initial spectrum (amplitude or tilt), the recombination history, the absolute Thomson normalization, or any likelihood statement. Those remain the gates tracked in issues #330, #522, #374, and #363 respectively. No statement in this note reads Planck data.

2 Branch inputs

Assumption 2.1 (A-GEOM: Einstein branch). *The observer-facing normal form carries a Lorentzian metric g_{ab} satisfying $G_{ab} + \Lambda g_{ab} = 8\pi G_{\text{geom}} \langle T_{ab} \rangle$, with $G_{\text{geom}} = \ell_\star^2$ from the edge-entropy/area-law theorem and $\Lambda \ell_\star^2 = 3\pi/N_\star$ from capacity closure. Provenance: branch theorem (paper stack A22/A5); not re-proved here.*

Assumption 2.2 (A-QED: recovered electromagnetic sector). *The realized branch carries the Standard Model quotient with a protected massless photon and a U(1) vertex of strength $\alpha = \alpha_{\text{root}}$ (source-side). Charged-lepton ratios are branch outputs; the absolute electron mass m_e is empirically anchored. Provenance: branch theorem (A20/A21) plus one declared import (m_e).*

Assumption 2.3 (A-BG: background). *The homogeneous limit of the branch is spatially flat FRW, $ds^2 = a^2(\tau) [-d\tau^2 + \delta_{ij}dx^i dx^j]$, with Friedmann dynamics from Assumption 2.1. Scalar perturbations are written in the conformal Newtonian gauge,*

$$ds^2 = a^2(\tau) \left[-(1 + 2\psi) d\tau^2 + (1 - 2\phi) \delta_{ij} dx^i dx^j \right],$$

with conventions of Ma–Bertschinger (1995). Overdots are ∂_τ .

Assumption 2.4 (A-REC: recombination import). *The free-electron history $n_e(\tau)$ (equivalently $x_e(\tau)$) is computed from standard atomic physics (Saha/Peebles-class) using imported atomic data. This is the same import class as the cesium-clock wall in the gravity stack: it contains no CMB-target information, so it is a permissible declared input under the anti-fitting rule, but it is an import, not a branch output.*

Assumption 2.5 (A-INIT: initial conditions (gated)). *Initial curvature perturbations $\zeta(\mathbf{k})$ are Gaussian with spectrum $\Delta_\zeta^2(k) = A_s (k/k_*)^{n_s-1}$. The Gaussian and statistically isotropic character is the MaxEnt screen-covariance theorem; the tilt certificate is issue #522; the amplitude and the radial screen-to- k lift are issue #330. This note treats A_s , n_s as gated inputs and derives nothing about them.*

3 Theorem B1: reversible repair gives Liouville transport

Definition 3.1 (One-particle record density). *Let $\mathcal{E}_\gamma$ be the photon-class excitation sector of the patch bundle. Between record commits, a single excitation is a record continued across patch overlaps. Coarse-graining patch position and overlap momentum labels defines the phase-space record density $f(x^i, q, \hat{n}, \tau)$ on the forward mass shell, where q is comoving momentum magnitude and $\hat{n}$ the direction.*

Theorem 3.2 (Refinement Liouville). *On a branch passing the simulation test, with (i) the protected massless photon row and overlap Lorentz kinematics (A-QED, A-GEOM), and (ii) repair moves organized in reversible write/check pairs (the orientation doubling in $m_{\text{rep}} = 2 \dim(\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)) = 24$), the between-commit record flow satisfies, in the refinement limit,*

$$\frac{Df}{d\tau} \equiv \frac{\partial f}{\partial \tau} + \frac{dx^i}{d\tau} \frac{\partial f}{\partial x^i} + \frac{dq}{d\tau} \frac{\partial f}{\partial q} + \frac{d\hat{n}}{d\tau} \cdot \frac{\partial f}{\partial \hat{n}} = C[f],$$

where the characteristics are the null geodesics of g_{ab} and $C[f]$ is supported on record commits only. In particular the collisionless flow preserves the Liouville measure.

Proof sketch, with the open receipt named. Characteristics. A single-excitation record continued across an overlap must agree with both patches’ readouts; by overlap Lorentz kinematics the unique continuation consistent with all overlapping patches is parallel transport along the branch metric’s null geodesic. Any other continuation produces a nonzero overlap mismatch, which the repair fixed point sets to zero; schedule independence (confluence) makes the continuation unique and schedule-free. Hence $dx/d\tau$, $dq/d\tau$, $d\hat{n}/d\tau$ are the geodesic-flow coefficients; on the background of Assumption 2.3, $dq/d\tau = q[\dot{\phi} - \dots]$ takes the standard form used in §5.

Measure preservation. Each repair move is a write/check pair: a bijection on the finite record configuration space with an explicit inverse (this is what the orientation doubling tracks). A finite composition of bijections has unit Jacobian on the induced finite phase space, for every schedule; confluence makes the endpoint schedule-independent, so the coarse-grained flow inherits

unit Jacobian. The refinement limit of a sequence of unit-Jacobian flows converging to the geodesic flow preserves the Liouville measure.

Open receipt. The convergence step — that the finite patch flow’s coarse-graining converges to the geodesic flow with vanishing dispersion — is exactly the kind of statement the abstract Lean layer proves in its setting (approximate schedule independence, refinement naturality) but has not been instantiated for the photon sector. We name it BOLTZ-R1 and give a simulator test in §7. Until BOLTZ-R1 is discharged, Theorem 3.2 is theorem-conditional. $\square$

Remark 3.3. The physical content is that *irreversibility enters only at commits*. Free streaming is unitary bookkeeping; entropy production is localized in the record-commit events that the collision functional counts. This is the transport-level face of the general OPH position that measurement-like irreversibility lives at the record interface.

4 Theorem B2: record commits give the Thomson collision functional

Theorem 4.1 (Commit-collision functional). *Let commits between photon-class and electron-class records occur with the soft-limit matrix element of the recovered $U(1)$ vertex (A -QED). Then, to first order in perturbations and at low occupancy, the collision functional in Theorem 3.2 is the Thomson functional: in the electron rest frame,*

$$C[f](\hat{n}) = a n_e \sigma_T \left[\frac{3}{16\pi} \int d\Omega' (1 + (\hat{n} \cdot \hat{n}')^2) f(\hat{n}') - f(\hat{n}) \right], \quad \sigma_T = \frac{8\pi}{3} \frac{\alpha^2}{m_e^2},$$

boosted to the frame where baryons move with velocity v_b . Moreover the write/check reversibility of commits implies detailed balance of the commit kernel, so the unique isotropic stationary point of C at fixed photon number is the Planck occupancy: the blackbody spectrum is the repair equilibrium of the record dynamics.

Proof sketch. At low occupancy the commit rate between momentum cells is bilinear in the occupancies with kernel given by the squared vertex; the soft (photon energy $\ll m_e$) limit of the recovered $U(1)$ vertex is classical Thomson scattering with the stated total cross-section and $1 + \cos^2$ angular kernel; this step is standard QED evaluated on the branch couplings and is not repeated here. Detailed balance: a write/check pair and its orientation reverse implement the commit and its inverse with equal kernel weight; a symmetric kernel yields an H -theorem for the commit dynamics, and the stationary condition at fixed number is the Bose equilibrium occupancy. Momentum conservation per commit, summed over commits, gives equal and opposite momentum transfer densities to the electron–baryon fluid, with coefficient fixed by $\bar{\rho}_\gamma/\bar{\rho}_b$ bookkeeping; this is the Compton drag used in §5. $\square$

Remark 4.2 (Provenance of the normalization). $\alpha = \alpha_{\text{root}}$ is source-side. m_e is an import (Assumption 2.2): OPH currently derives charged-family *ratios*, not the absolute scale, so σ_T ’s normalization is import-tagged (BOLTZ-N1 in the ledger). $n_e(\tau)$ is import-tagged by Assumption 2.4 (BOLTZ-X1).

Remark 4.3 (OPH-specific channels). Two and only two places exist where OPH kernels can modify this functional at linear order: (i) a number-changing repair-exchange channel $\Gamma_{\text{rec}}(k, a)$ adds a monopole source and a spectral-distortion term — it enters $C[f]$ additively and is gated by the #374 receipts (active fiber, physical clock, common-parent response); (ii) a collar-packet stress $\rho_A, B_A(k, a)$ enters only the gravitational source side (the Einstein constraints in §5), never $C[f]$. With both gates closed at zero, the transport below is exactly standard.

5 Theorem B3: the linear multipole hierarchy

Theorem 5.1 (Reduction to the standard hierarchy). *Linearize Theorems 3.2–4.1 on the background of Assumption 2.3 with scalar perturbations, expand the temperature brightness $F_\gamma(\mathbf{k}, \hat{n}, \tau)$ and polarization brightness G_γ in Legendre multipoles with $\mu = \hat{k} \cdot \hat{n}$. Writing $\delta_\gamma = F_{\gamma 0}$, $\theta_\gamma = \frac{3}{4}kF_{\gamma 1}$, $\sigma_\gamma = \frac{1}{2}F_{\gamma 2}$, and $\dot{\kappa} \equiv an_e\sigma_T$, the system is (Ma–Bertschinger 1995, eq. 63):*

$$\begin{aligned}\dot{\delta}_\gamma &= -\frac{4}{3}\theta_\gamma + 4\dot{\phi}, \\ \dot{\theta}_\gamma &= k^2\left(\frac{1}{4}\delta_\gamma - \sigma_\gamma\right) + k^2\psi + \dot{\kappa}(\theta_b - \theta_\gamma), \\ 2\dot{\sigma}_\gamma &= \frac{8}{15}\theta_\gamma - \frac{3}{5}kF_{\gamma 3} - \frac{9}{5}\dot{\kappa}\sigma_\gamma + \frac{1}{10}\dot{\kappa}(G_{\gamma 0} + G_{\gamma 2}), \\ \dot{F}_{\gamma\ell} &= \frac{k}{2\ell+1}\left[\ell F_{\gamma(\ell-1)} - (\ell+1)F_{\gamma(\ell+1)}\right] - \dot{\kappa}F_{\gamma\ell}, \quad \ell \geq 3, \\ \dot{G}_{\gamma\ell} &= \frac{k}{2\ell+1}\left[\ell G_{\gamma(\ell-1)} - (\ell+1)G_{\gamma(\ell+1)}\right] + \dot{\kappa}\left[-G_{\gamma\ell} + \frac{1}{2}(F_{\gamma 2} + G_{\gamma 0} + G_{\gamma 2})\left(\delta_{\ell 0} + \frac{\delta_{\ell 2}}{5}\right)\right],\end{aligned}$$

coupled to baryons,

$$\dot{\delta}_b = -\theta_b + 3\dot{\phi}, \quad \dot{\theta}_b = -\frac{\dot{a}}{a}\theta_b + c_b^2k^2\delta_b + k^2\psi + \frac{4\bar{\rho}_\gamma}{3\bar{\rho}_b}\dot{\kappa}(\theta_\gamma - \theta_b),$$

and to the Einstein constraints of Assumption 2.1,

$$k^2\phi + 3\frac{\dot{a}}{a}\left(\dot{\phi} + \frac{\dot{a}}{a}\psi\right) = -4\pi Ga^2\delta\rho_{\text{tot}}, \quad k^2(\phi - \psi) = 12\pi Ga^2(\bar{\rho} + \bar{p})\sigma_{\text{tot}},$$

where $\delta\rho_{\text{tot}}$ and σ_{tot} include the gated ρ_A, B_A contributions of Remark 4.3 (zero until their receipts pass). With $\Gamma_{\text{rec}} = B_A = 0$ this is verbatim the standard Einstein–Boltzmann system.

Proof. Mechanical: insert the geodesic characteristics of Theorem 3.2 on the perturbed metric, the collision functional of Theorem 4.1, keep first order, and project on Legendre multipoles. Every step is the textbook computation (Ma–Bertschinger 1995; Dodelson, *Modern Cosmology*); no OPH-specific term survives except the two gated channels, which we have displayed at their unique insertion points. $\square$

6 Acoustic normal form, damping, and the line of sight

Proposition 6.1 (Acoustic oscillations). *In the tight-coupling regime $\dot{\kappa} \gg k, \frac{\dot{a}}{a}$, define $R \equiv 3\bar{\rho}_b/4\bar{\rho}_\gamma$ and $\Theta_0 = \delta_\gamma/4$. To leading order in $1/\dot{\kappa}$,*

$$\frac{d}{d\tau}\left[(1+R)\dot{\Theta}_0\right] + \frac{k^2}{3}\Theta_0 = -\frac{k^2}{3}(1+R)\psi + \frac{d}{d\tau}\left[(1+R)\dot{\phi}\right],$$

a driven oscillator with sound speed $c_s^2 = 1/[3(1+R)]$. The WKB solution gives, at last scattering τ_* ,

$$[\Theta_0 + \psi](\tau_*) \propto \cos(kr_s(\tau_*)), \quad r_s(\tau_*) = \int_0^{\tau_*} c_s d\tau,$$

so extrema sit at $k_n r_s(\tau_*) = n\pi$: compressions (odd n) are baryon-enhanced, rarefactions (even n) suppressed, and the dark-matter driving of the potentials boosts the third peak — every feature of the measured TT curve, produced by this system and not by any OPH-specific term.

Proposition 6.2 (Silk damping). *At next order in $1/\kappa$ the oscillator acquires the diffusion envelope e^{-k^2/k_D^2} with*

$$k_D^{-2} = \int_0^\tau \frac{d\tau'}{6(1+R)\dot{\kappa}} \left[\frac{R^2}{1+R} + \frac{16}{15} \right],$$

the 16/15 arising from the polarization terms of Theorem 5.1; photon diffusion between commits is the physical mechanism, and its OPH reading is transparent: on scales crossed by a random walk of commit-free transport legs, the record field cannot maintain contrast.

Proposition 6.3 (Line-of-sight solution). *With visibility $g(\tau) = \dot{\kappa} e^{-\kappa(\tau, \tau_0)}$, the temperature multipoles today are*

$$\Theta_\ell(k, \tau_0) = \int_0^{\tau_0} d\tau \left\{ g \left[\Theta_0 + \psi + \frac{\Pi}{4} + \frac{3}{4k^2} \ddot{\Pi} \right] j_\ell(k(\tau_0 - \tau)) + \left(\frac{g v_b}{k} \right) \dot{j}_\ell + e^{-\kappa} (\dot{\psi} + \dot{\phi}) j_\ell \right\},$$

($\Pi = F_{\gamma 2} + G_{\gamma 0} + G_{\gamma 2}$), and

$$C_\ell^{TT} = 4\pi \int \frac{dk}{k} \Delta_\zeta^2(k) \left| \frac{\Theta_\ell(k, \tau_0)}{\zeta(k)} \right|^2.$$

Given Assumption 2.5's gated (A_s, n_s) and the imports of Assumptions 2.2–2.4, this reproduces the curve computed by the diagnostic CAMB lane; agreement of an independent implementation with CAMB at the sub-percent level is receipt BOLTZ-V1.

7 Receipts and simulator tests

Receipt	Statement	Status
BOLTZ-R1	Finite-lattice refinement Liouville: coarse-grained photon-class record transport between commits converges to geodesic flow, schedule-independently, with vanishing measure defect under refinement.	open; simulator test below
BOLTZ-N1	σ_T normalization: absolute m_e is a declared import until a charged-family absolute-scale theorem exists.	declared import
BOLTZ-X1	$x_e(\tau)$ recombination history from imported atomic data; contains no CMB-target information.	declared import
BOLTZ-V1	Independent hierarchy implementation matches CAMB TT/TE/EE at stated tolerances with identical inputs.	open; engineering
Tilt certificate	Theorem-grade n_s (clock normalization).	issue #522
Amplitude + lift	A_s and radial screen-to- k map.	issue #330
Kernels	$\Gamma_{\text{rec}}, \rho_A, B_A$ physical receipts.	issue #374
Likelihood	Frozen likelihood protocol and hashes.	issue #363

Simulator test for Boltz-R1. Inject a labeled low-weight record packet on the 64k lattice; propagate under N random repair schedules; measure the arrival-time and dispersion statistics

of the packet across the screen. Pass condition: schedule-to-schedule divergence of the packet trajectory consistent with zero at the run’s mismatch tolerance, and dispersion shrinking under refinement (16k→64k→256k) at the rate the refinement-naturality bounds predict. Fail condition: schedule-dependent transport, which would falsify Theorem 3.2 on the implemented branch.

8 What this note does and does not close

It closes, conditionally on the cited branch theorems, the *form* of the transport: OPH inherits the standard Einstein–Boltzmann system, with exactly two gated insertion points for new kernels and three tagged imports. Consequently the physical-CMB prediction gate no longer contains “derive the Boltzmann equations” as an open item; it contains BOLTZ-R1 and BOLTZ-V1 plus the pre-existing gates (#330, #522, #374, #363). It does not certify initial conditions, normalizations, recombination, or likelihoods, and nothing here reads Planck data. Under the anti-fitting rule, the acoustic peak structure remains attributable to standard plasma physics; OPH’s falsifiable exposure stays concentrated in the primordial inputs, exactly where issues #330 and #522 put it.

References

- C.-P. Ma and E. Bertschinger, *Cosmological Perturbation Theory in the Synchronous and Conformal Newtonian Gauges*, ApJ 455, 7 (1995).
- S. Dodelson and F. Schmidt, *Modern Cosmology*, 2nd ed., Academic Press (2020).
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- W. Hu and M. White, *CMB Anisotropies: Total Angular Momentum Method*, Phys. Rev. D 56, 596 (1997).
- OPH paper stack: *Observers Are All You Need*; *Reality as a Consensus Protocol*; *Recovering Relativity and the Standard Model from Observer Overlap Consistency*; *OPH Cosmology Finite-Source CMB Program* (this repository).